CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 4: Linear Inequations (In One Variable)

Topic-wise short notes with solved examples

An inequation compares two expressions using a sign such as less than or greater than. This chapter covers solving linear inequations and showing their solutions on a number line.

1 Inequations and their signs

- A **linear inequation** in one variable x has one of the forms $ax + b < c$, $ax + b > c$, $ax + b \leq c$ or $ax + b \geq c$, with $a \neq 0$.

Sign	Meaning
$<$	is less than
$>$	is greater than
$\leq$	is less than or equal to
$\geq$	is greater than or equal to

- The **replacement set** (domain) is the set from which the values of x are taken: N (natural numbers), W (whole numbers), Z or I (integers) or R (real numbers).
- The **solution set** is the set of all values from the replacement set that satisfy the inequation.

2 Rules for solving

- **Rule 1**: the same number may be added to or subtracted from both sides. The sign stays the same.
- **Rule 2**: both sides may be multiplied or divided by the same **positive** number. The sign stays the same.
- **Rule 3**: if both sides are multiplied or divided by the same **negative** number, the sign **reverses**.
- **Rule 4**: if the two sides are interchanged, the sign reverses: $a < b$ means $b > a$.
- A term can be moved to the other side with its sign changed, exactly as in an equation.

Example 2.1 Solve $3x - 5 \leq 7$, $x \in N$.

SOLUTION

$3x \leq 12$, so $x \leq 4$

The natural numbers not greater than 4 are 1, 2, 3, 4.

Solution set $= \{1, 2, 3, 4\}$

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Example 2.2 Solve $-3x + 4 > 16, x \in Z$.

SOLUTION

$-3x > 12$. Dividing by -3 reverses the sign: $x < -4$

Solution set $= \{\ldots, -7, -6, -5\}$

Example 2.3 Solve $\frac{x}{2} - 1 \geq \frac{x}{3} - 2, x \in R$.

SOLUTION

Multiply both sides by 6: $3x - 6 \geq 2x - 12$

$3x - 2x \geq -12 + 6$

$x \geq -6$; **solution set** $= \{x : x \geq -6,\ x \in R\}$

REMEMBER

To avoid a negative coefficient, move the x terms to the side where the coefficient comes out positive.

3 Solution set and the replacement set

- The same inequation has different solution sets for different replacement sets.

Inequation	Replacement set	Solution set
$x < 3$	N	$\{1, 2\}$
$x < 3$	W	$\{0, 1, 2\}$
$x < 3$	Z	$\{\ldots, 0, 1, 2\}$
$x < 3$	R	All real numbers less than 3

Example 3.1 Find the largest integer x for which $2x + 7 < 20$.

SOLUTION

$2x < 13$, so $x < 6.5$

The largest integer is 6.

4 Representing the solution on a number line

- For N, W or Z: mark each solution with a **thick dot**. Use three dots beyond the last one to show that the solutions continue.
- For R: draw a **thick line** over the part of the number line that forms the solution.
- At an end value: use a **filled (dark) circle** if the value is included ($\leq$ or $\geq$) and a **hollow circle** if it is not included ($<$ or $>$).
- An arrow at one end shows that the solution continues without end in that direction.

Example 4.1 Describe the number line representation of $-2 < x \leq 3, x \in R$.

SOLUTION

Mark a hollow circle at -2, because -2 is not included.

Mark a filled circle at 3, because 3 is included.

Draw a thick line joining the two circles.

5 Combining inequations

- A double inequation such as $a < px + q \leq b$ is two inequations joined by "and". Solve each part and take the values common to both.
- **"and"** means the **intersection** of the two solution sets. **"or"** means their **union**.

Example 5.1 Solve $-3 < 2x - 1 \leq 5, x \in R$.

SOLUTION

Add 1 throughout: $-2 < 2x \leq 6$

Divide by 2: $-1 < x \leq 3$

Solution set $= \{x : -1 < x \leq 3,\ x \in R\}$: a hollow circle at -1, a filled circle at 3 and a thick line between them.

Example 5.2 Given $A = \{x : 5x - 3 > 3x + 5,\ x \in R\}$ and $B = \{x : 2x - 7 \leq 9,\ x \in R\}$, find $A \cap B$.

SOLUTION

A: $2x > 8$, so $x > 4$

B: $2x \leq 16$, so $x \leq 8$

$A \cap B = \{x : 4 < x \leq 8,\ x \in R\}$

Example 5.3 Solve $2x - 5 \leq 5x + 4 < 11, x \in I$.

SOLUTION

Left part: $2x - 5 \leq 5x + 4$ gives $-9 \leq 3x$, so $x \geq -3$

Right part: $5x + 4 < 11$ gives $5x < 7$, so $x < 1.4$

Together: $-3 \leq x < 1.4$

Solution set $= \{-3, -2, -1, 0, 1\}$

6 Common mistakes

- Forgetting to reverse the sign when multiplying or dividing by a negative number.
- Ignoring the replacement set: listing fractions when $x \in N$, or listing a few numbers when $x \in R$.
- Using a filled circle for $<$ or $>$.
- Solving only one half of a double inequation.

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